CHAPTER 1 LINEAR EQUATIONS AND LINEAR TRANSFORMATIONS

1.1 System of Linear Equations

- A **linear equation** in the variables x_1, x_2, ..., x_n is an equation that can be written in the form

$$a_1x_1 + a_2x_2 + \ldots + a_nx_n = b$$

where b and the coefficients a_1, a_2, ..., a_n are real or complex numbers.

If $b = 0$, we said that the system is **homogeneous**. Otherwise, we said that the system is **inhomogeneous**.

- **Example**
  - $4x_1 - 3x_2 = x_1 - 2$
  - $x_2 = 2\left(\sqrt{6} - x_1\right) + x_3$
  - $4x_1 - 5x_2 = x_1x_2$
  - $x_1 = \sqrt{x_2} - 3$

- A **system of linear equations** (or **linear system**) is a collection of one or more linear equations involving the same variables.

$$\begin{cases} 2x_1 - x_2 + 1.5x_3 = 8 \\ x_1 - 4x_3 = -7 \end{cases}$$

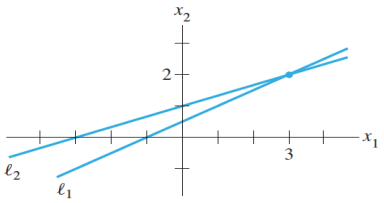
- A **solution** of the system is a list $(s_1, s_2, \ldots, s_n)$ of numbers satisfy each equation in the system when the values s_1, s_2, …, s_n are substituted for x_1, x_2, …, x_n respectively.

- The set of all possible solutions is called the **solution set** of the linear system.

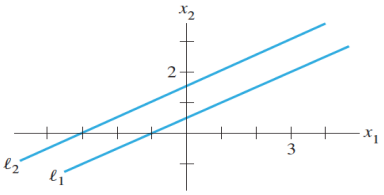
- Two linear systems are called **equivalent** if they have the same solution set.

- **Example** Solve the following linear systems.

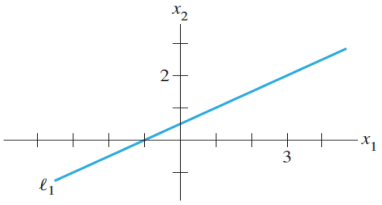
(a)
$$\begin{cases} x_1 - 2x_2 = -1 \\ -x_1 + 3x_2 = 3 \end{cases}$$



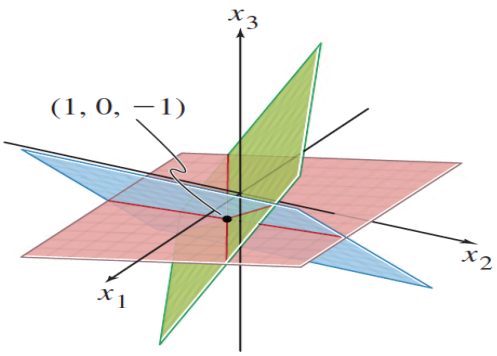
(b)
$$\begin{cases} x_1 - 2x_2 = -1 \\ -x_1 + 2x_2 = 3 \end{cases}$$

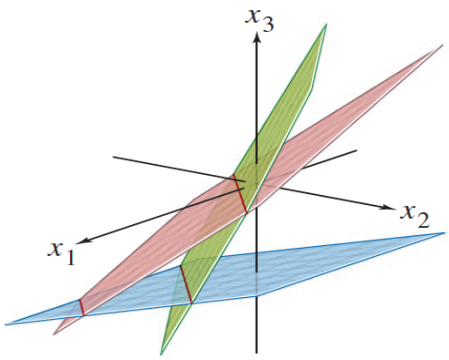


(c)
$$\begin{cases} x_1 - 2x_2 = -1 \\ -x_1 + 2x_2 = 1 \end{cases}$$



- **Example** A solution of a three by three system of equations (three equations and three unknowns) can be viewed as intersection of three plans. Once again, the system can be one unique solution (left figure, ignore the labelling $(1, 0, -1)$), infinite many solution (image you open a book that only has three pages), or no solution (right figure).





- **Solutions of Linear Systems**

A linear system can have

- No solution
- Exactly one solution
- Infinitely many solutions

- The essential information of a linear system can be recorded in a rectangular array called a **matrix**.

- **Example** Find the coefficient matrix and the augmented matrix of the following linear system.

$$\begin{cases} x_1 + x_2 + 2x_3 = 9 \\ 2x_1 + 4x_2 - 3x_3 = 1 \\ 3x_1 + 6x_2 - 5x_3 = 0 \end{cases}$$

- **Example** Solve the following linear system.

$$\begin{cases} x_1 + x_2 + 2x_3 = 9 \\ 2x_1 + 4x_2 - 3x_3 = 1 \\ 3x_1 + 6x_2 - 5x_3 = 0 \end{cases}$$

- **Elementary Row Operations**

• Interchange two rows. • Multiply all entries in a row by a nonzero constant. • Replace one row by the sum of itself and a multiple of another row.

- Two matrices are called **row equivalent** if there is a sequence of elementary row operations that transforms one matrix into the other.

- **Theorem**

If the augmented matrices of two linear systems are row equivalent, then the two systems have the same solution set.

- A system of equations is said to be **consistent** if it has at least one solution. It is **inconsistent** if it has no solutions.

- **Example** Determine whether the following system is consistent.

$$\begin{cases} x_1 + x_2 + 2x_3 = 9 \\ 2x_1 + 4x_2 - 3x_3 = 1 \\ 3x_1 + 6x_2 - 5x_3 = 0 \end{cases}$$

- **Example** Determine whether the following system is consistent.

$$\begin{cases} \quad x_2 - \quad 4x_3 = \ 8 \\ 2x_1 - \ 3x_2 + \quad 2x_3 = \ 1 \\ 4x_1 - \ 8x_2 + \ 12x_3 = \ 1 \end{cases}$$

Exercises Section 1.1, Practice problems. Section 1.1, 3,7,9,11,13,15,21,25,27,29,31.

1.2 Row Reduction and Echelon Forms

- **Echelon form and reduced echelon form**

A rectangular matrix is in **echelon form** (or **row echelon form**) if it has the following three properties:

1. All nonzero rows are above any rows of zeros.
2. Each leading entry of a row is in a column to the right of the leading entry of the row above it.
3. All entries in a column below a leading entry are zeros.

If a matrix echelon form satisfies the following additional conditions, then it is in **reduced echelon form** (or **reduced row echelon form**)

4. The leading entry in each nonzero row is 1.
5. Each leading 1 is the only nonzero entry in its column.

EXAMPLE 1 The following matrices are in echelon form. The leading entries (■) may have any nonzero value; the starred entries (∗) may have any value (including zero).

$$\begin{bmatrix} \blacksquare & * & * & * \\ 0 & \blacksquare & * & * \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & \blacksquare & * & * & * & * & * & * & * & * \\ 0 & 0 & 0 & \blacksquare & * & * & * & * & * & * \\ 0 & 0 & 0 & 0 & \blacksquare & * & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & \blacksquare & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \blacksquare & * \end{bmatrix}$$

The following matrices are in reduced echelon form because the leading entries are 1's, and there are 0's below *and above* each leading 1.

$$\begin{bmatrix} 1 & 0 & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 & * & 0 & 0 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 1 & 0 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 1 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 0 & 1 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & * \end{bmatrix}$$

- **Theorem**

Each matrix is row equivalent to one and only one reduced echelon form.

- A **pivot position** in a matrix A is a location in A that corresponds to a leading 1 in the reduced echelon form of A. A **pivot column** is a column of A that contains a pivot position.
- The variables corresponding to pivot columns in the matrix are called the **basic variables**. Others are called **free variables**.

- **Example** Suppose that the augmented matrix of a linear system with unknowns x_1, x_2, x_3 and x_4 has been reduced to

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 3 \\ 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 5 \end{bmatrix}$$

solve the system.

- **Example** Suppose that the augmented matrix of a linear system with unknowns x_1, x_2, x_3 and x_4 has been reduced to

$$\left[\begin{array}{rrrrr} 1 & 2 & 0 & 4 & 3 \\ 0 & 1 & 0 & 9 & -1 \\ 0 & 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 0 & -2 \end{array}\right]$$

  solve the system.

- **Example** Suppose that the augmented matrix of a linear system with unknowns x_1, x_2, x_3 and x_4 has been reduced to

$$\left[\begin{array}{rrrrr} 1 & 2 & 0 & 4 & 3 \\ 0 & 1 & 0 & 9 & -1 \\ 0 & 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

  solve the system.

- **Example** Suppose that the augmented matrix of a linear system with unknowns x_1, x_2, x_3 and x_4 has been reduced to

$$\left[\begin{array}{ccccc} 1 & 2 & 3 & 4 & 3 \\ 0 & 0 & 1 & 9 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

solve the system.

- **Example** Solve the following linear system

$$\begin{cases} x_1 + 6x_2 + 2x_3 - 5x_4 - 2x_5 = -4 \\ \qquad\quad 2x_3 - 8x_4 - x_5 = 3 \\ \qquad\qquad\quad + x_5 = 7 \end{cases}$$

- **Example** Determine the existence and uniqueness of the solutions to the following linear system

(a)
$$\begin{cases} & 3x_2 - 6x_3 + 6x_4 + 3x_5 = -5 \\ 3x_1 - 7x_2 + 8x_3 - 5x_4 + 8x_5 = 9 \\ 3x_1 - 9x_2 + 12x_3 - 9x_4 + 6x_5 = 15 \end{cases}$$

(b)
$$\begin{cases} & 3x_2 - 6x_3 + 6x_4 + 3x_5 = -9 \\ 3x_1 - 7x_2 + 8x_3 - 5x_4 + 8x_5 = 9 \\ 3x_1 - 9x_2 + 12x_3 - 9x_4 + 6x_5 = 15 \end{cases}$$

(c)
$$\begin{cases} & 3x_2 - 6x_3 + 6x_4 + 3x_5 = 0 \\ 3x_1 - 7x_2 + 8x_3 - 5x_4 + 8x_5 = 0 \\ 3x_1 - 9x_2 + 12x_3 - 9x_4 + 6x_5 = 0 \end{cases}$$

- **Example** Solve the following linear system

$$\begin{cases} x_1 + 3x_2 - 2x_3 + 2x_5 = 0 \\ 2x_1 + 6x_2 - 5x_3 - 2x_4 + 4x_5 - 3x_6 = -1 \\ 5x_3 + 10x_4 + 15x_6 = 5 \\ 2x_1 + 6x_2 + 8x_4 + 4x_5 + 18x_6 = 6 \end{cases}$$

Exercises Section 1.2, Practice Problems. Section 1.2, 1,5,7,11,13,17,19,23,25,27,29,31.

1.3 Vectors

- A vector in $\mathbb{R}^2$ can be written as

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

- Given two vectors, we can **define** their **sum** (add corresponding entries) and **scalar multiplication** (multiply each entry).

- **Example** Given $\mathbf{u} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 2 \\ -5 \end{bmatrix}$, find $4\mathbf{u}$ and $4\mathbf{u} - 3\mathbf{v}$.

- A vector in $\mathbb{R}^3$ can be written as

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$$

- A vector in $\mathbb{R}^n$ can be written as

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$$

  The zero vector is denoted by $\mathbf{0}$.

- **Algebraic Properties**

  - $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
  - $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$
  - $\mathbf{u} + \mathbf{0} = \mathbf{0} + \mathbf{u} = \mathbf{u}$
  - $-\mathbf{u} = (-1)\mathbf{u}$
  - $c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$
  - $(c + d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$
  - $c(d\mathbf{u}) = cd\mathbf{u}$
  - $1\mathbf{u} = \mathbf{u}$

- Given vectors $\mathbf{v}_1$, $\mathbf{v}_2$, ..., $\mathbf{v}_n$ in $\mathbb{R}^n$, a linear combination of $\mathbf{v}_1$, $\mathbf{v}_2$, ..., $\mathbf{v}_n$ is

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \ldots + c_n\mathbf{v}_n$$

where c_1, c_2, ..., c_n are scalars.

- **Example** Let $\mathbf{a}_1 = \begin{bmatrix} 1 \\ -2 \\ -5 \end{bmatrix}$, $\mathbf{a}_2 = \begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix}$, determine whether $\mathbf{b} = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}$ and $\mathbf{c} = \begin{bmatrix} 2 \\ 4 \\ -3 \end{bmatrix}$ can be written as linear combinations of $\mathbf{a}_1$ and $\mathbf{a}_2$.

- A vector equation

$$x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \ldots + x_n\mathbf{a}_n = \mathbf{b}$$

has the same solution set as the linear system whose augmented matrix is

$$\begin{bmatrix}\mathbf{a}_1 & \mathbf{a}_2 & \ldots & \mathbf{a}_n & \mathbf{b}\end{bmatrix}$$

In particular, $\mathbf{b}$ can be generated as a linear combination of $\mathbf{a}_1$, $\mathbf{a}_2$, …, $\mathbf{a}_n$ if and only if there exists a solution to the linear system corresponding to the matrix above.

- **Definition**

If $\mathbf{v}_1$, $\mathbf{v}_2$, …, $\mathbf{v}_p$ are in $\mathbb{R}^n$, then the set of all linear combinations of $\mathbf{v}_1$, $\mathbf{v}_2$, …, $\mathbf{v}_p$ is denoted by $\text{span}\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ and is called the subset of $\mathbb{R}^n$ spanned by $\mathbf{v}_1$, $\mathbf{v}_2$, …, $\mathbf{v}_p$.

Exercises Section 1.3, Practice Problems. Section 1.3, 5,7,9,11,13,17,21,25,33.

2.1 Matrix Operations

- An $m \times n$ matrix can be written as $A = [a_{ij}]$

$$= \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1j} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2j} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{i1} & a_{i2} & \dots & a_{ij} & \dots & a_{in} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mj} & \dots & a_{mn} \end{bmatrix}.$$

  where a_{ij} denotes the entry at the ith row and jth column. It can also be written as

$$A = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \dots & \mathbf{a}_n \end{bmatrix}$$

- A **diagonal matrix** is a square matrix whose non-diagonal entries are zeros.

- The identity matrix I_n.

- The zero matrix.

- If A and B are $m \times n$ matrices, then the sum $A + B$ is **defined** to be the $m \times n$ matrix whose entries are the summation of corresponding entries of A and B.

- If r is a scalar and A is a matrix, then by **definition**, rA is simply obtained by multiplying every entry of A by r.

- **Example** Let $A = \begin{bmatrix} 4 & 0 & 5 \\ -1 & 3 & 2 \end{bmatrix}$, $B = \begin{bmatrix} -2 & 1 & 3 \\ -4 & 2 & -1 \end{bmatrix}$, find $2A - 3B$.

- **Theorem**

  - $A + B = B + A$
  - $(A + B) + C = A + (B + C)$
  - $A + 0 = A$
  - $r(A + B) = rA + rB$
  - $(r + s)A = rA + sA$
  - $r(sA) = (rs)A$

- **Matrix Multiplication**

  If A is an $m \times n$ matrix, B is an $n \times p$ matrix, then the product C is an $m \times p$ matrix where

$$c_{ij} = \begin{bmatrix} a_{i1} & a_{i2} & \dots & a_{in} \end{bmatrix} \begin{bmatrix} b_{1j} \\ b_{2j} \\ \vdots \\ b_{nj} \end{bmatrix} = a_{i1}b_{1j} + a_{i2}b_{2j} + \ldots + a_{in}b_{nj}$$

- **Example** Let $A = \begin{bmatrix} 3 & 4 & 5 \\ -2 & -1 & 7 \end{bmatrix}$ and $B = \begin{bmatrix} -2 & 3 \\ 1 & -5 \\ -3 & -1 \end{bmatrix}$. Find AB and BA.

- **Example** Let $A = \begin{bmatrix} 3 & 4 & 5 \\ -2 & -1 & 7 \\ 6 & -7 & 8 \\ 0 & -2 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} -2 & 3 & 2 \\ 1 & -5 & 0 \\ -3 & -1 & 1 \end{bmatrix}$. Find the third row of AB.

- **Theorem**

  - $A(BC) = (AB)C$
  - $A(B + C) = AB + AC$
  - $(B + C)A = BA + CA$
  - $r(AB) = (rA)B = A(rB)$
  - $I_m A = A = AI_n$

- **Example** Let $A = \begin{bmatrix} 3 & 4 & 5 \\ -2 & -1 & 7 \\ 0 & 2 & -3 \end{bmatrix}$. Find A^3.

- The **transpose** of a $m \times n$ matrix A is a $n \times m$ matrix where

$$(A^T)_{ij} = A_{ji}$$

- **Example** Let $A = \begin{bmatrix} 3 & 4 & 5 \\ -2 & -1 & 7 \end{bmatrix}$ and $B = \begin{bmatrix} -2 & 3 \\ 1 & -5 \\ -3 & -1 \end{bmatrix}$. Find $A^T B^T$ and $(BA)^T$.

- **Theorem**

  - $(A^T)^T = A$
  - $(A + B)^T = A^T + B^T$
  - $(rA)^T = rA^T$
  - $(AB)^T = B^T A^T$

Exercises Section 2.1 Exercises, 1,3,5,7,9,27.

1.4 The Matrix Equation $A\mathbf{x} = \mathbf{b}$

- A linear system

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \ldots + a_{2n}x_n = b_2 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \ldots + a_{mn}x_n = b_m \end{cases}$$

can be written in the form

$$A\mathbf{x} = \mathbf{b}$$

where

$$A = \begin{bmatrix} a_{11} & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} & \ldots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \ldots & a_{mn} \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

$$\mathbf{a}_1 = \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix}, \mathbf{a}_2 = \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix}, \ldots \mathbf{a}_n = \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix}$$

are called the columns of the matrix A, and the linear system is equivalent to the vector equation

$$x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \ldots + x_n\mathbf{a}_n = \mathbf{b}$$

Hence, the equation $A\mathbf{x} = \mathbf{b}$ has a solution if and only if $\mathbf{b}$ is a linear combination of the columns of A.

- **Example** Let $A = \begin{bmatrix} 1 & 3 & 4 \\ -4 & 2 & -6 \\ -3 & -2 & -7 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$. Is the equation $A\mathbf{x} = \mathbf{b}$ consistent for all possible b_1, b_2 and b_3?

- **Theorem**

Let A be an $m \times n$ matrix. The followings are equivalent:

(a) For each $\mathbf{b} \in \mathbb{R}^m$, the equation $A\mathbf{x} = \mathbf{b}$ has a solution.

(b) Each $\mathbf{b} \in \mathbb{R}^m$ is a linear combination of the columns of A.

(c) The columns of A span $\mathbb{R}^m$.

(d) A has a pivot position in every row.

- **Note**

For every $\mathbf{b} \in \mathbb{R}^m$, the equation $A\mathbf{x} = \mathbf{b}$ has a solution if and only if the row echelon form of A does not have a row consisting entirely of zeros.

Exercises Section 1.4, Practice Problems. Page 56, 1,3,5,7,11,13,15,17,19,21,25.

1.5 Solution Sets of Linear Systems

- **Example** Determine whether the following homogeneous system has a non-trivial solution. Write the solutions in parametric vector form.

$$\begin{cases} 3x_1 + 5x_2 - 4x_3 = 0 \\ -3x_1 - 2x_2 + 4x_3 = 0 \\ 6x_1 + x_2 - 8x_3 = 0 \end{cases}$$

- **Example** Describe the solution set of the following equation

$$10x_1 - 3x_2 - 2x_3 = 0$$

Write the solutions in parametric vector form.

- **Example** Describe the solutions of $A\mathbf{x} = \mathbf{b}$, where

$$A = \begin{bmatrix} 3 & 5 & -4 \\ -3 & -2 & 4 \\ 6 & 1 & -8 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 7 \\ -1 \\ -4 \end{bmatrix}$$

Write the solutions in parametric vector form.

- **Theorem**

> If the equation $A\mathbf{x} = \mathbf{b}$ has a solution given by $\mathbf{x} = \mathbf{p}$, then the solution set of the equation $A\mathbf{x} = \mathbf{b}$ is the set of all vectors of the form $\mathbf{x} = \mathbf{v}_h + \mathbf{p}$, where $\mathbf{v}_h$ is any solution of the homogeneous equation $A\mathbf{x} = \mathbf{0}$.

Exercises Section 1.5, Practice Problems. Section 1.5, 1,3,5,7,9,11,13,15,29,31,33,35.

1.7 Linear Independence

- **Linearly Independent and Linearly Dependent**

A set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_p\}$ in $\mathbb{R}^n$ is said to be **linearly independent** if

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \ldots + c_p\mathbf{v}_p = \mathbf{0}$$

implies

$$c_1 = c_2 = \ldots = c_p = 0.$$

The set is said to be **linearly dependent** if there exists $c_1, c_2, \ldots, c_p$ not all zero such that

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \ldots + c_p\mathbf{v}_p = \mathbf{0}$$

- **Example** If the set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_p\}$ in $\mathbb{R}^n$ contains the zero vector $\mathbf{0}$, then it is linearly dependent.

- **Example** Let $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$ and $\mathbf{v}_3 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$. Determine if the set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is linearly independent. If not, find a linear relation among $\mathbf{v}_1$, $\mathbf{v}_2$ and $\mathbf{v}_3$.

- **Example** Determine if the columns of the matrix $A = \begin{bmatrix} 0 & 1 & 4 \\ 1 & 2 & -1 \\ 5 & 8 & 0 \end{bmatrix}$ are linearly independent.

- **Theorem**

> The columns of a matrix A are linearly independent if and only if the equation $A\mathbf{x} = \mathbf{0}$ has only the trivial solution, if and only if every column of A is pivot.

- **Example** Determine whether the following sets of vectors are linearly independent.

  (a) $\mathbf{v}_1 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 6 \\ 2 \end{bmatrix}$

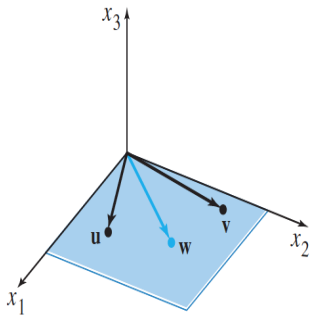
  (b) $\mathbf{v}_1 = \begin{bmatrix} 3 \\ 2 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 6 \\ 2 \end{bmatrix}$

- **Theorem**

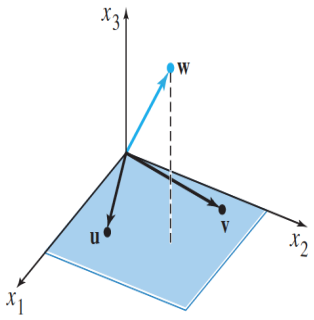
> A set of two vectors $\{\mathbf{v}_1, \mathbf{v}_2\}$ is linearly dependent if and only if one of the vectors is a multiple of the other. The set is linearly independent if and only if neither of the vectors is a multiple of the other.

- **Theorem**

> An indexed set $S = \{\mathbf{v}_1, ..., \mathbf{v}_p\}$ of two or more vectors is linearly dependent if and only if at least one of the vectors in S is a linear combination of the others.



Linearly dependent,
w in Span{**u**, **v**}



Linearly independent,
w *not* in Span{**u**, **v**}

- **Theorem**

> If $p > n$, then the set $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_p\}$ of vectors in $\mathbb{R}^n$ is linearly dependent.

- **Example** Determine by inspection if the given set is linearly dependent.

(a) $\begin{bmatrix} 1 \\ 7 \\ 6 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 9 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ 5 \end{bmatrix}, \begin{bmatrix} 4 \\ 1 \\ 8 \end{bmatrix}$

(b) $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 9 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ 5 \end{bmatrix}$

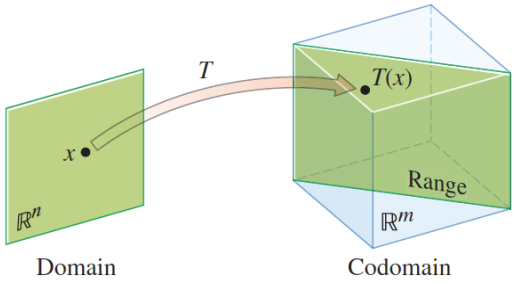
(c) $\begin{bmatrix} 2 \\ 0 \\ 9 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ 5 \end{bmatrix}$

(d) $\begin{bmatrix} -2 \\ 4 \\ 6 \\ 10 \end{bmatrix}, \begin{bmatrix} 3 \\ -6 \\ -9 \\ -15 \end{bmatrix}$

Exercises Section 1.7 Exercises, 1,3,5,7,13,15,17,19,27.

1.8 Linear Transformations

- A **transformation** T from $\mathbb{R}^n$ to $\mathbb{R}^m$ is a rule that assigns to each vector $\mathbf{x}$ in $\mathbb{R}^n$ a vector $T(\mathbf{x})$ in $\mathbb{R}^m$. $\mathbb{R}^n$ is called the domain of T and $\mathbb{R}^m$ is called the codomain of T. $T(\mathbf{x})$ is called the image of $\mathbf{x}$. The set of all images $T(\mathbf{x})$ is called the range of T.



- **Example** Let $A = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix}$, $\mathbf{u} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} 3 \\ 2 \\ -5 \end{bmatrix}$, $\mathbf{c} = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}$, and define a transformation $T : \mathbb{R}^2 \to \mathbb{R}^3$ by $T(\mathbf{x}) = A\mathbf{x}$.

  (a) Find $T(\mathbf{u})$.

  (b) Find an $\mathbf{x}$ in $\mathbf{R}^2$ whose image under T is $\mathbf{b}$. Is there more than one such $\mathbf{x}$?

  (c) Determine whether $\mathbf{c}$ is in the range of T.

- **Linear Transformations**

A transformation T is linear if

(a) $T(\mathbf{u}+\mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$

(b) $T(c\mathbf{u}) = cT(\mathbf{u})$

- (**For your information only, if you ever heard that linear transform maps straight lines to straight lines and if you want to know how to represent a straight line in** $\mathbb{R}^n$) Suppose $\mathbf{u}$ and $\mathbf{v}$ are two points in $\mathbb{R}^n$. It is not difficult to see that as t runs through $-\infty$ to ∞, $t\mathbf{u} + (1-t)\mathbf{v}$ runs through all points lying on the straight line passing through $\mathbf{u}$ and $\mathbf{v}$. For example, in $\mathbb{R}^2$, we know any point $\begin{bmatrix} x \\ y \end{bmatrix}$ lying on the straight line passing through $\mathbf{u} = \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} x_2 \\ y_2 \end{bmatrix}$ satisfies $y = \frac{y_1-y_2}{x_1-x_2}(x-x_2) + y_2$. Hence $\frac{y-y_2}{x-x_2} = \frac{y_1-y_2}{x_1-x_2}$. If $x - x_2 = t(x_1 - x_2)$, then $y - y_2 = t(y_1 - y_2)$. It means that if we set $t = \frac{x-x_2}{x_1-x_2}$, then $\begin{bmatrix} x \\ y \end{bmatrix} = t \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} + (1-t) \begin{bmatrix} x_2 \\ y_2 \end{bmatrix}$.

  Question: if t is required to lie between 0 and 1, what is $t\mathbf{u} + (1-t)\mathbf{v}$?

  You can think of t as time and think of $t\mathbf{u} + (1-t)\mathbf{v} \in \mathbb{R}^n$ as the position of a particle at time t. Question: If you have learned multivariable calculus, can you find what is the velocity and acceleration of this particle?

  Under the transformation T, this line is mapped to $T(t\mathbf{u}+(1-t)\mathbf{v}) = tT(\mathbf{u})+(1-t)T(\mathbf{v})$, which is again a line that passes through two points $T(\mathbf{u})$ and $T(\mathbf{v})$. Hence linear transformation maps straight lines to straight lines.

- **Theorem**

Matrix transformations are linear transformations.

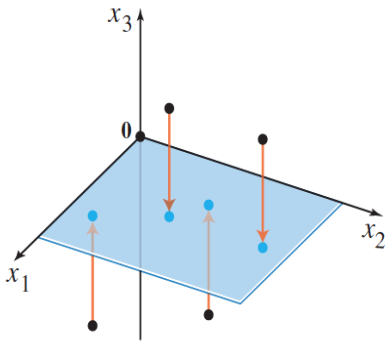
- **Theorem**

If T is a linear transformation,

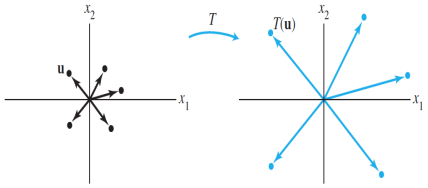
(a) $T(\mathbf{0}) = \mathbf{0}$

(b) $T(c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \ldots + c_p\mathbf{v}_p) = c_1T(\mathbf{v}_1) + c_2T(\mathbf{v}_2) + \ldots + c_pT(\mathbf{v}_p)$

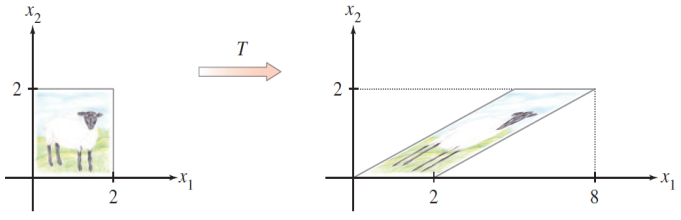
- **Example** Describe the linear transformation $\mathbf{x} \mapsto A\mathbf{x}$, where $\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$.



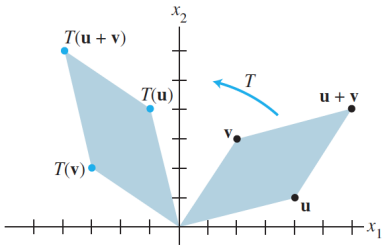
- **Example** Describe the linear transformation $\mathbf{x} \mapsto A\mathbf{x}$, where $\mathbf{A} = \begin{bmatrix} r & 0 \\ 0 & r \end{bmatrix}$.



- **Example** Describe the linear transformation $\mathbf{x} \mapsto A\mathbf{x}$, where $\mathbf{A} = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$.



- **Example** Describe the linear transformation $\mathbf{x} \mapsto A\mathbf{x}$, where $\mathbf{A} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$.



- **Example** Describe the linear transformation $\mathbf{x} \mapsto A\mathbf{x}$, where $\mathbf{A} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$.

Exercises Section 1.8 Exercises, 1,3,5,7,9,11,15,17,19.

1.9 The Matrix of a Linear Transformations

- **Example** Let $T : \mathbb{R}^2 \to \mathbb{R}^3$ be a linear transformation such that

$$T\left(\begin{bmatrix}1\\0\end{bmatrix}\right) = \begin{bmatrix}5\\-7\\2\end{bmatrix}, \quad T\left(\begin{bmatrix}0\\1\end{bmatrix}\right) = \begin{bmatrix}-3\\8\\0\end{bmatrix}$$

  Find $T(\mathbf{x})$ for any $\mathbf{x} \in \mathbb{R}^2$.

- In $\mathbf{R}^n$, we define

$$\mathbf{e}_1 = \begin{bmatrix}1\\0\\0\\\vdots\\0\end{bmatrix}, \mathbf{e}_2 = \begin{bmatrix}0\\1\\0\\\vdots\\0\end{bmatrix}, \mathbf{e}_3 = \begin{bmatrix}0\\0\\1\\\vdots\\0\end{bmatrix}, \ldots, \mathbf{e}_n = \begin{bmatrix}0\\0\\0\\\vdots\\1\end{bmatrix}$$

- **Theorem**

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation. There exists a unique $m \times n$ matrix A such that

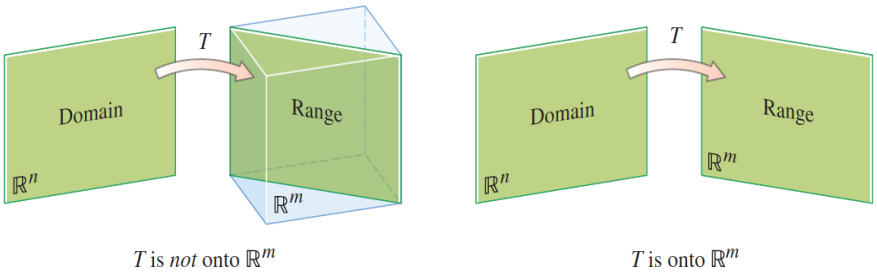
$$T(\mathbf{x}) = A\mathbf{x}$$

In fact,

$$A = \begin{bmatrix} T(\mathbf{e}_1) & T(\mathbf{e}_2) & \dots & T(\mathbf{e}_n) \end{bmatrix}$$

- **Definition**

A mapping $T : \mathbb{R}^n \to \mathbb{R}^m$ is said to be **onto** $\mathbb{R}^m$ if each $\mathbf{b}$ in $\mathbb{R}^m$ is the image of *at least one* $\mathbf{x}$ in $\mathbb{R}^n$.



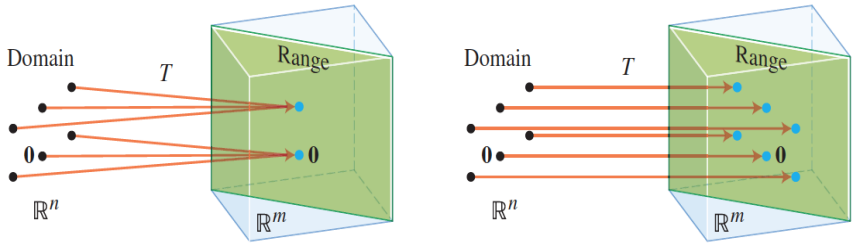
T is *not* onto $\mathbb{R}^m$ — *T* is onto $\mathbb{R}^m$

- **Definition**

A mapping $T : \mathbb{R}^n \to \mathbb{R}^m$ is said to be **one-to-one** if each $\mathbf{b}$ in $\mathbb{R}^m$ is the image of *at most one* $\mathbf{x}$ in $\mathbb{R}^n$.

Equivalently, T is one-to-one if, for each $\mathbf{b}$ in $\mathbb{R}^m$, the equation $T(\mathbf{x}) = \mathbf{b}$ has either a unique solution or none at all.

The mapping T is *not* one-to-one when some $b \in \mathbb{R}^m$ is the image of more than one vector in $\mathbb{R}^n$. If there is no such $\mathbf{b}$, then T is one-to-one.



T is *not* one-to-one — *T* is one-to-one

- **Example** Let T be the linear transformation whose standard matrix is

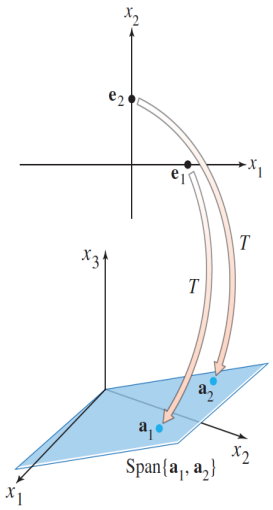
$$A = \begin{bmatrix} 1 & -4 & 8 & 1 \\ 0 & 2 & -1 & 3 \\ 0 & 0 & 0 & 5 \end{bmatrix}$$

Does T maps $\mathbb{R}^4$ onto $\mathbb{R}^3$? If T a one-to-one mapping?

- **Theorem**

> Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation. Then T is one-to-one if and only if the equation $T(\mathbf{x}) = \mathbf{0}$ has only the trivial solution.

- **Example** Let $T(x_1, x_2) = (3x_1 + x_2, 5x_1 + 7x_2, x_1 + 3x_2)$. Show that T is one-to-one. Does T maps $\mathbb{R}^2$ onto $\mathbb{R}^3$?



The transformation T is not onto $\mathbb{R}^3$.

Exercises Read Tables 1-4 of Section 1.9. Section 1.9, 1,3,5,11,17,19,21,25,27.